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IOP Publishing Measurement Science and Technology
Meas. Sci. Technol. **27** (2016) 065301 (12pp) doi:10.1088/0957-0233/27/6/065301

Automated system for measuring the surface dilational modulus of liquid–air interfaces

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Received 11 January 2016, revised 22 March 2016
Accepted for publication 6 April 2016
Published 27 April 2016



Abstract

The surface dilational modulus is a crucial parameter for describing the rheological properties of aqueous surfactant solutions. These properties are important for many technological processes. The present paper describes a fully automated instrument based on the oscillating bubble technique. It works in the frequency range from 1 Hz to 500 Hz, where surfactant exchange dynamics governs the relaxation process. The originality of instrument design is the consistent combination of modern measurement technologies with advanced imaging and signal processing algorithms. Key steps on the way to reliable and precise measurements are the excitation of harmonic oscillation of the bubble, phase sensitive evaluation of the pressure response, adjustment and maintenance of the bubble shape to half sphere geometry for compensation of thermal drifts, contour tracing of the bubbles video images, removal of noise and artefacts within the image for improving the reliability of the measurement, and, in particular, a complex trigger scheme for the measurement of the oscillation amplitude, which may vary with frequency as a result of resonances. The corresponding automation and programming tasks are described in detail. Various programming strategies, such as the use of MATLAB® software and native C++ code are discussed. An advance in the measurement technique is demonstrated by a fully automated measurement. The instrument has the potential to mature into a standard technique in the fields of colloid and interface chemistry and provides a significant extension of the frequency range to established competing techniques and state-of-the-art devices based on the same measurement principle.

Keywords: surface rheology, dilational modulus, automated measurements, image processing, signal processing, graphical user interfaces, oscillating bubble technique

(Some figures may appear in colour only in the online journal)

1. Introduction

The surface dilational modulus E is a fundamental system parameter of aqueous surfactant solutions which play a decisive role in the physics of foams and emulsions, in coating and spraying processes, the field of microfluidics, cleaning processes or biological systems (e.g. the oxygen exchange in the pulmonary alveoli) [1]. The E-modulus describes the ability of the fluid to restore the surface tension at an instance of stress.

Amphiphiles combine two opposing structural elements in a single molecule: a polar or ionic headgroup and a hydrophobic tail. As a result of the prevailing molecular asymmetry, amphiphiles spontaneously adsorb at air-water and oil-water

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interfaces thereby reducing the surface tension and altering surface rheology. This class of substances is widely used for tuning the interfacial properties of fluids.

An expansion or compression of the surface area leads to the deviation of the surface coverage from the equilibrium. As a result, relaxation processes are initiated to restore the equilibrium value within the adsorption layer. Different relaxation processes govern the system behavior over different time-scales. The surface coverage Γ depends on the bulk concentration c of the surfactant. It is accessible from the equilibrium surface tension isotherm $\gamma(c)$. Assuming harmonic oscillation of the surface area A and the surface tension γ with time t, i.e. the corresponding perturbations ΔA and $\Delta\gamma$ being proportional to $\exp(j\omega t)$, the surface dilational modulus E is a complex quantity defined as

$$E(\omega, c) = |E(\omega, c)| \cdot \exp\left[j\phi(\omega, c)\right] = -\frac{\Delta\underline{\gamma}}{\Delta\underline{A}}A, \qquad (1)$$

where j is the imaginary unit, ω is the angular oscillation frequency, A is the surface area, ϕ is the phase difference angle, $\Delta\underline{\gamma}$ and $\Delta\underline{A}$ are the phasors of the dynamic surface tension and the surface area, respectively.

Since different relaxation processes are dominant on characteristic time-scales, several experimental techniques are employed to follow the time evolution of the system. A comprehensive review of the experimental techniques can be found in [1]. At frequencies below 0,1 Hz a Langmuir balance equipped with oscillating barriers can be used [2, 3]. Here, the surface area of the liquid is changed by movable barriers and the surface tension is measured by a Wilhelmy system. Higher frequencies can be reached by drop shape tensiometry. The shape of a drop is given by the balance of gravity and surface tension. The video analysis of the oscillating drop yields the E-modulus via the Gauss Laplace equation. The upper frequency limit of this technique is about 5 Hz as discussed in [4].

The important mid-frequency range from 1 Hz to 500 Hz can be investigated by means of oscillating bubble tensiometry [5, 6]. In this technique, a bubble is formed at the tip of a capillary which is immersed into the liquid and forced to harmonic oscillations, e.g. by a piezoelectric transducer. Simultaneous analysis of the amplitude and phase of the resulting pressure response with respect to the induced oscillation of the surface area yields the complex surface dilational E-modulus.

The mathematical framework for measuring the dilational rheological properties of gas-liquid interfaces by analyzing small amplitude radial oscillations of an air bubble in a liquid phase has been introduced in [7]. Hitherto, only a few works emphasized the measurement techniques and the associated instrumentation. Russev *et al* described an instrument, which works with small drops and bubbles, and measures interfacial tension and dilational rheological properties for frequencies up to 100 Hz [8]. Ravera *et al* suggested to take advantage of digital CCD cameras for reliable measurements of dilational visco-elasticity at higher frequencies [9]. A commercially available instrument is specified at frequencies from 1 to 100 Hz to determine the dilational elasticity and viscosity of interfacial layers [10].

Although the oscillating bubble method is known for more than 30 years, reliable data of dynamic surface properties at frequencies above 100 Hz are scarce. This can be attributed to experimental challenges and the necessity to take into account hydrodynamic factors [1, 11].

Capillary wave techniques are considered to be an alternative for measuring the dynamic properties in the frequency range up to 1000 Hz. So far, also this technique has not emerged as a standardized routine characterization of aqueous interfaces due to experimental difficulties and the complexity of data interpretation [12].

For frequencies exceeding 100 Hz, an advance in the measurement techniques is desired in order to promote further progress in colloid and interface science [13]. The purpose of this paper is to address this experimental challenge and to outline the programming tasks in the development of a reliable oscillating-bubble apparatus and the implementation of an automation of measurements. The presented device has the potential to mature into a standard technique in colloid and surface chemistry. The automation concept and the proposed technical solutions could be of interest for engineers and scientists developing complex automated measurement systems combining imaging and signal processing techniques with the methods of control engineering.

The paper is organized as follows. Section 2 presents the experimental setup, whereas the apparatus, real-time signal and imaging processing algorithms are described in detail. The design takes advantage of modern hard- and software solutions. Illustrative experimental results, demonstrating the capability of the measurement system, are presented and discussed in section 3. Conclusions are drawn in the final section 4.

2. Experimental setup

2.1. Measurement apparatus

Figure 1 shows a schematic cross sectional view of the oscillating bubble device.

The central part of the apparatus is the measuring chamber (outer dimensions: $4 \times 4 \times 4$ cm^3) manufactured from stainless steel. It contains the sample volume of approximately 4 ml. The two piezoelectric actuators, the pressure sensor, the capillary, the spyholes, the in- and outlet ports are connected to the chamber. To ensure a stable bubble below the capillary, the container must be sealed tightly. All chamber openings and contacts have been caulked by using different kinds of sealing rings. To eliminate undesired bubbles in the chamber during the filling process, a specifically designed tilting mechanism (not shown in figure 1) is implemented. It inclines the chamber towards the vertical axis in such a way that unwanted air bubbles may escape through the provided outlet (not shown in figure 1).

The principle of the oscillating bubble method requires the formation of a hemispherical bubble attached to the tip of a capillary (here: diameter of about 0,3 mm). For proper measurements of surface rheology, the capillary has to meet specific requirements. Stability of the three phase contact line

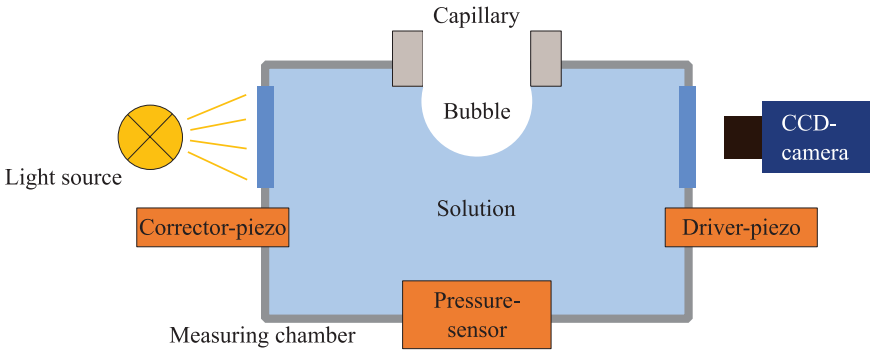


Figure 1. Schematic cross sectional view of the oscillating bubble device. Please note the drawing is not to scale, the size of the bubble is enlarged.

between the capillary, the aqueous solution and the air has to be assured by an appropriate wetting regime [14–16].

To form the bubble at the tip of the capillary, a piezoelectric micrometer screw (P-853.00, Physik Instrumente GmbH, Germany) is used. It is attached to the chamber and its piston (referred to as corrector-piezo in figure 1) tunes the sample volume finely. The size of the bubble can be adjusted in two ways. A rough adjustment is performed by the micrometer screw integrated into the piezoelectric actuator. To ensure the hemispherical shape of the bubble at the reference state, a more precise regulation of the bubble size is achieved by using the piezoelectric control range of this actuator.

The piezoelectric driver is controlled by the data acquisition USB device from Jaeger Computergesteuerte Messtechnik GmbH (ADwin-light-16). The ADwin-system is equipped with a signal processor. Therefore, the data processing tasks are decoupled from the PC and its current workload. The ADwin-system, comprising analogue and digital inputs and outputs, is responsible for fast and accurate acquisition of measurement data [17]. Furthermore, it is used as the signal generator for the excitation of periodic surface area perturbations. An external voltage amplifier is required to drive the piezoelectric actuator in its specified voltage range.

The second piezoelectric actuator (P-830.40, Physik Instrumente GmbH, Germany, referred to as driver-piezo in figure 1) forces the bubble to harmonic sinusoidal oscillations. It features a response time in the microsecond range and subnanometer resolution. Due to the incompressibility of the fluid, the motion of the piezoelectric driving piston determines the bubble size. The sinusoidal excitation voltage trace is generated by the ADwin-system.

Forced periodic oscillations of the hemispherical bubble lead to changes of the pressure across the curved liquid-vapor interface [7]. The highly sensitive pressure transducer (Type 7261, Kistler Instrumente GmbH, Germany [18]) monitors the pressure variations Δp inside the measuring chamber caused by the varying bubble radius and surface area [19]. The output signal of the pressure sensor is converted into electrical voltage by a charge amplifier with femto-Coulumb accuracy (Elektronik Manufaktur Mahlsdorf, Germany). This voltage is digitized by the ADwin-system. Evaluating the pressure response and the corresponding changes in surface area by means of image processing allows one to investigate the amplitude of the surface dilational modulus $|E|$ as well as its phase angle ϕ between the induced bubble deformation and pressure response [20].

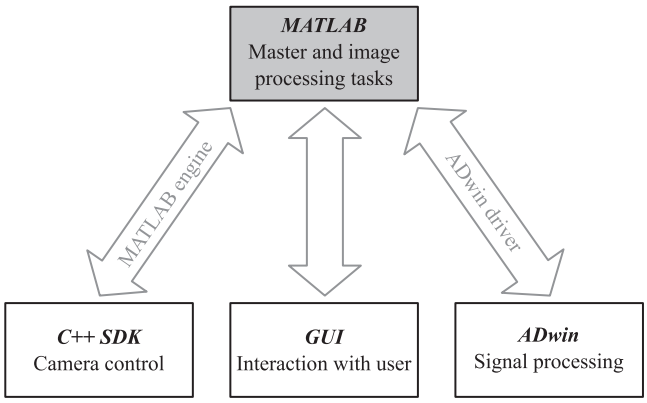


Figure 2. Interaction between MATLAB, the ADwin-system and the CCD camera.

According to (1), determination of the surface dilational modulus E requires the evaluation of the surface area changes of the bubble during its oscillation. Therefore, a CCD-camera (UK1117 series, ehd imaging GmbH, Germany) is attached to a microscope (MZM, Mikroskop Technik Rathenow, Germany) and used to observe the bubble at the tip of the capillary as well as to take images of the bubble at its maximum and minimum extensions. This is achieved by triggering the camera by an external signal generated by the ADwin-system. Therefore, the imaging of fast bubble oscillations is accomplished in a stroboscopic fashion. To ensure high reproducibility and accuracy of the measurements, the images recorded by the camera are evaluated for verification of the hemispherical shape of the bubble. The CCD-camera is powered via USB and the data transfer is performed via the same interface. The camera takes gray-scale images with a resolution of 8 bits per pixel. The integrated DSP allows one to perform the image pre-processing directly within the camera. The high quality 1/2 inch CCD-sensor provides a high signal-to-noise ratio and the global shutter allows capturing of images of fast moving objects. The exposure time can be reduced to a minimum of 50 μs at a maximum frame rate of 25 frames per second [21].

The illumination required for taking images of the quickly moving bubble, oscillating with frequencies up to 500 Hz,

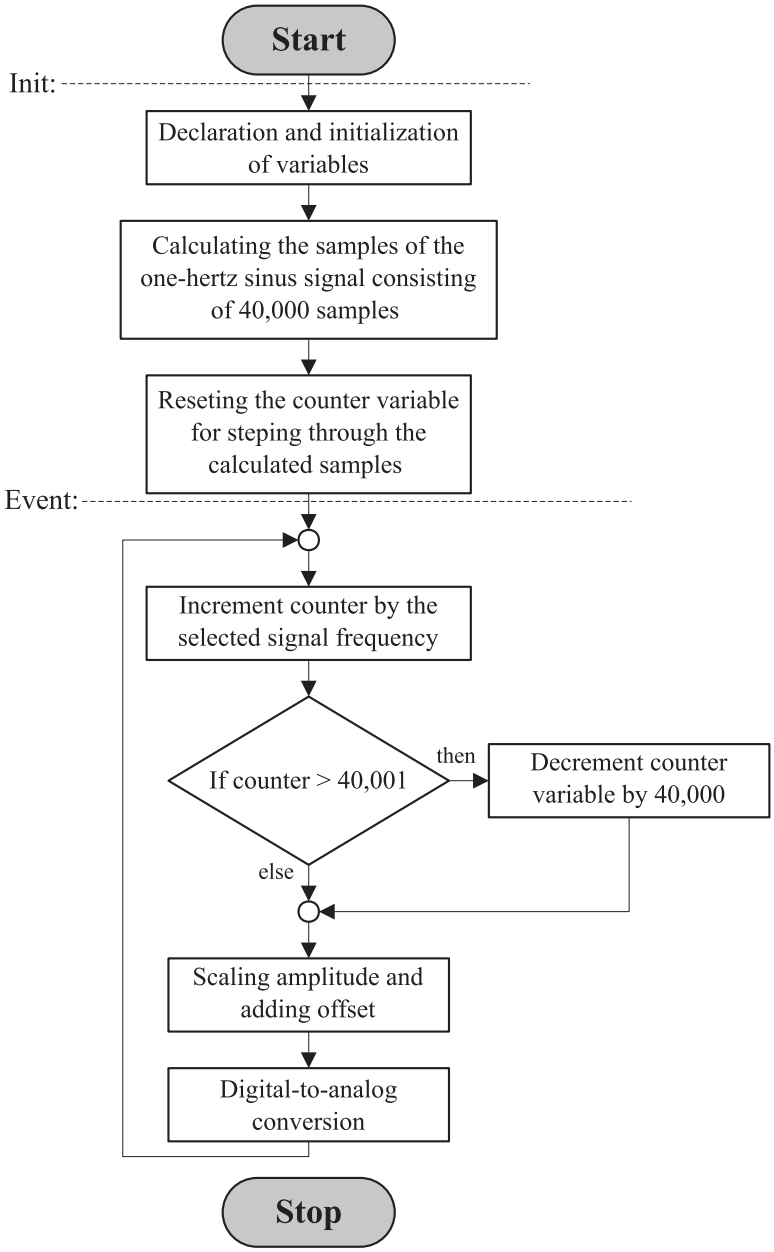


Figure 3. Generating the sinusoidal signal to drive the piezoelectric piston and to set the bubble in oscillations.

is provided by a high power LED (ALUSTAR 3 W, ledxon GmbH, Germany) enabling to set the exposure time to the minimal possible value. The CCD-camera observes the bubble via a spyhole from the opposite side. As the exposure time of 50 μs is much smaller than the minimum bubble oscillation period of 2 ms, it is ensured that bubble dimensions do not change significantly during exposure and the images show the bubble at its extreme positions.

The apparatus is mounted on an anti-vibration table to decouple external influences from the measurement. As a result of the mismatch between the bubble size and the sample volume, the size of the bubble is extremely sensitive to thermal drifts. To ensure the hemispherical shape of the bubble at the tip of the capillary, an automated regulation of the bubble size has been developed.

All user defined parameters can be entered via a graphical user interface (GUI) programmed in MATLAB[3]. It includes the camera's live image, display of all electrical signals and a visualization of the results. The measured data sets are stored as ASCII-files. The automation of the measurement procedure including digital image processing tasks, as well as the setting of camera parameters are based on MATLAB scripts. Figure 2 illustrates how interfaces allow access from MATLAB to the ADwin-system and the CCD-camera.

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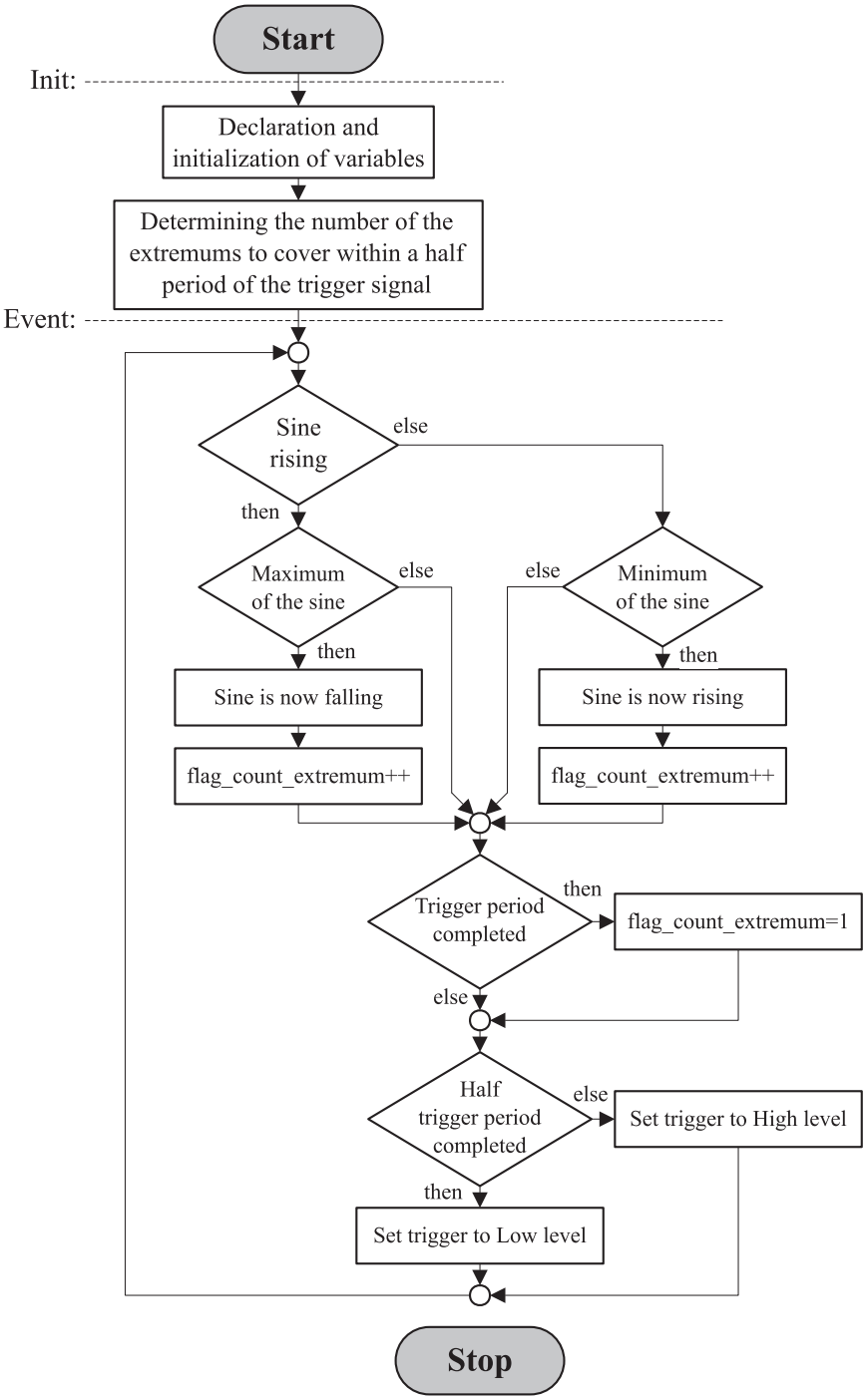


Figure 4. Generation of the trigger-signal for the CCD-camera.

2.2. Signal processing

The two piezoelectric actuators, the pressure sensor, the CCD-camera and the DC-motor for the tilting mechanism of the chamber are connected with the ADwin-light-16-USB-device which is equipped with a 32 bit signal processor and enables data acquisition and automation task under real-time conditions. The system comes with an Integrated Development Environment (IDE) allowing the design of user defined programs using ADBasic as the programming language. The fundamental structure of the ADBasic-programmes consists of an initialization routine which is executed only once at each start of the process, and a functional section referred to as EVENT which is executed periodically with fixed and reliable timing corresponding to an endless loop [22].

Due to the variety of functions the ADwin-system must fulfill, the program structure is organized as a finite state machine, comprising the following states:

- Initialization,
- Generating the excitation signal for the bubble oscillation and the trigger signal for the CCD-camera together with the pressure sensor signal processing,
- Automated control of the bubble size,

- Manual adjustment of the bubble size,
- Tilting of the chamber during its filling and
- Waiting state.

In the following, the main functionality of several states is explained in more detail.

2.2.1. Excitation signal for the bubble oscillation. The piezoelectric driving piston, referred to as driver-piezo in figure 1, forces the bubble to periodic oscillations. Therefore, it must be driven by a sinusoidal voltage trace. This signal is generated by the ADwin-system. The analogue output of the ADwin-system is clocked at 40 kHz. Therefore, a one-Hertz sinusoidal oscillation is built from 40 000 samples. The signals for the generation of frequencies of n-fold multiples of the one Hertz oscillation are generated by picking the value of only every nth sampling point and issuing at the original time interval slots. The functional principle using the ADwin-system is shown in figure 3.

This method of generating a periodic signal has two main advantages as compared to a numerical calculation of the sinusoidal signal:

- The ADwin-system executes the calculation of the one-Hertz sine signal within its initialization routine. As there is no sine calculation in the main program, the processor workload is reduced.
- Calculating a sinusoidal function involves numerical inaccuracies. As the periodic output signal is a sequence of various periods, the connection between two single waves may not be exact and may lead to phase noise.

2.2.2. Trigger for the CCD-camera. A microscope equipped with a CCD-camera takes sequential images at maximum and minimum bubble extensions. These states are reached at the maximum and minimum values of the sinusoidal excitation signal for the driver-piezo, respectively. The CCD-camera is triggered by a TTL-signal generated by the ADwin-system in conjunction with the sinusoidal driving voltage, whereas the acquisition of an image is initiated by rising edge of a pulse. The maximum frame rate of the camera is limited to about 25 pictures per second. Since the frequency of the excitation signal ranges from 1 to 500 Hz, the camera is not capable of obtaining images at every maximum and minimum bubble extensions above a certain limiting frequency. To overcome this problem, the trigger for the camera is generated from the sinusoidal driving signal by counting its extremal points and simultaneously setting the trigger frequency as high as possible. The implementation of the camera trigger generation is shown in figure 4 and the resulting output signal of the ADwin-system is displayed in figure 5.

2.2.3. Evaluating the signal from the pressure sensor. The signal measured by the pressure sensor is first amplified by the external charge amplifier and subsequently digitized by the ADwin board. The peak-to-peak value of the digitized pressure response is proportional to the magnitude of the surface dilational modulus E. The phase shift ϕ between the

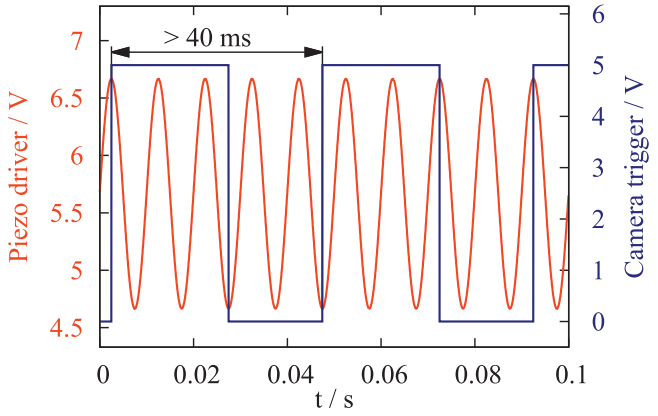


Figure 5. Oscilloscope screen showing the 100 Hz sinusoidal signal for the bubble oscillation (orange) and the trigger signal for the CCD-camera (blue).

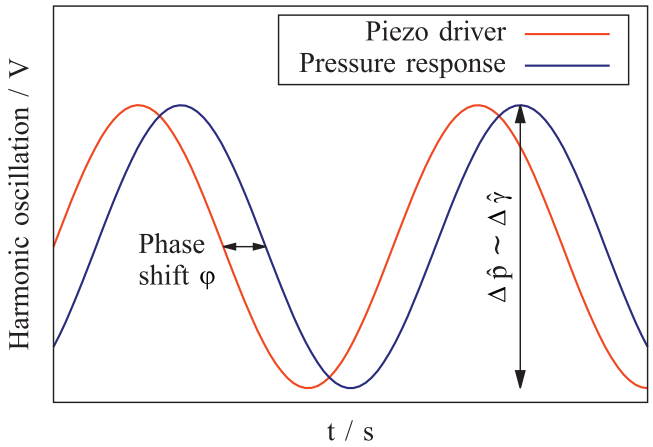


Figure 6. Identifying the relevant parameters for the surface rheology of fluids from the pressure response.

sinusoidal driving signal and the pressure response is the second characteristic of the complex modulus and schematically shown in figure 6.

These quantities are computed by the ADwin-system using the Discrete Fourier Transformation (DFT). The first Fourier coefficients a_1 and b_1 , the phase angle ϕ and the amplitude of the measured signal $\Delta\hat{p}$ are given by

$$a_1 = \sum_{k=0}^{N-1} u(k \cdot T_{\text{sam}}) \cdot \cos\left(\frac{2\pi}{N} \cdot k\right), \tag{2}$$

$$b_1 = \sum_{k=0}^{N-1} u(k \cdot T_{\text{sam}}) \cdot \sin\left(\frac{2\pi}{N} \cdot k\right), \tag{3}$$

$$\phi = \arctan\left(\frac{-a_1}{b_1}\right) \text{ and} \tag{4}$$

$$\Delta\hat{p} = \frac{1}{N} \cdot \sqrt{{a_1}^2 + {b_1}^2}, \tag{5}$$

where N is the total number of samples and T_{sam} is the sampling period [23]. By using the preprocessed sine function of

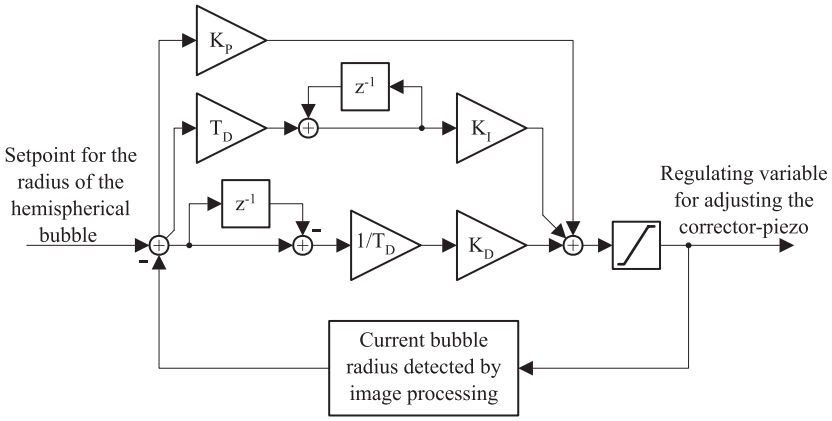


Figure 7. Signal flow diagram of the realized digital PID-controller. Tuning of the proportional coefficient K_P, the integral coefficient K_I and the derivative coefficient K_D permits the adaptation of the controller to the given specific needs. The time period for a processing step, i.e. evaluating an image and returning a new value for the current bubble size is given by the parameter T_D.

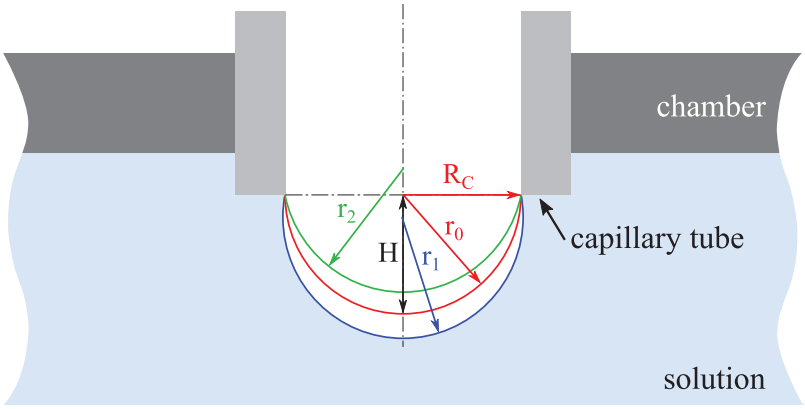


Figure 8. Schematic cross sectional view of the bubble oscillations.

the driving voltage and the fact that a cosine is a shifted sine, it was possible to reduce the processing effort.

2.2.4. Automated stabilization of the bubble size. The chosen diameter of the capillary is a compromise between the amplitude of the pressure response $\Delta\hat{p}$ and the sensitivity to thermal drifts of bubble volume. A smaller bubble maximizes the pressure response but also increases the sensitivity to thermal drifts. The miniaturization of the chamber volume is determined by the size of the pressure sensor, whose choice was based on a market survey, and the physical limitations in manufacturing of key elements of the measurement setup. However, it is more challenging to maintain a small bubble at constant volume due to its higher sensitivity towards thermal drifts. In order to overcome this problem, a proportional-integral-derivative(PID)-controller is used to ensure a hemispherical reference shape of the bubble shape at the tip of the capillary. Figure 7 shows the general mode of operation of the controller implemented in the ADwin-system. An image processing scheme (described later in this article) determines the setpoint value of the ideal hemispherical bubble, while a continuously executed algorithm investigates the current size of the bubble. The mentioned controller compares the setpoint with the actual value and drives the corrector-piezo (see figure 1) according to the calculated deviations.

The parameters K_P, K_I and K_D have been adjusted to the experimental requirements in the course of several experimental studies to guarantee both a smooth approach to the hemispherical reference shape and the prevention of overshoots. The coefficient T_D is measured by a MATLAB routine every time the most recently obtained image has been evaluated.

2.3. Image processing

Images of the oscillating bubble and the capillary are analyzed by a MATLAB script. Figure 8 shows the cross sectional view of the oscillating bubble at its hemispherical reference shape (red) and its maximum (blue) and minimum (green) extensions. In the following, the image processing for determining the height of the bubble below the capillary is described. This height at the hemispherical shape is used as the setpoint for the PID-controller.

2.3.1. Image preprocessing. The elimination of noise and possible inhomogeneities (e.g. dirt or dust particles from the surrounding air on the lens) in the captured image is a prerequisite for further image processing steps [24]. Edge detection algorithms work more effectively in the absence of such imperfections. The main result of preprocessing is the filling

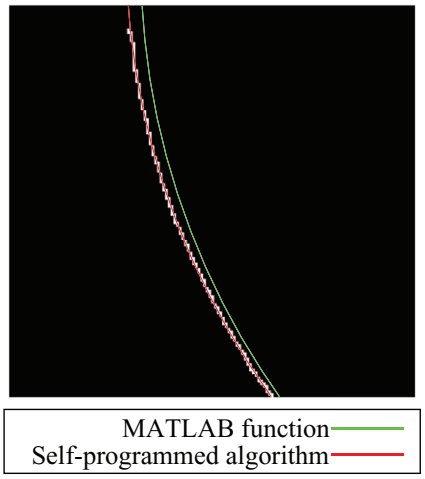


Figure 9. Comparison of the built-in MATLAB function for detecting circles with the self-programmed algorithm. The binary image in the background is the result of the edge detection scheme by Canny.

of 'gaps' in the image without changing the size of the larger structures [25]. The effects of dilation followed by erosion are enhanced by applying the operations twice with the same structuring element.

The described morphological operations are implemented with functions from the MATLAB Image Processing Toolbox. The MATLAB version used is 8.5.0.197613 (R2015a).

2.3.2. Edge detection. An edge is defined as an abrupt change in the intensity or pixel value of an image [25]. Such intensity changes occur along the transition line between the bubble and the surrounding solution, as well as between the capillary and the solution. The detected edge pixels are required as input for contour and shape detection algorithms, as they reduce the amount of data to the essential structural information [26].

The MATLAB Image Processing Toolbox allows detecting of edges using different methods [27]. Comparison of different methods in terms of computing times and accuracy lead to the conclusion, that the Canny edge detector [28] works most effectively in our case. Although it is almost three times slower than alternative algorithms, it has three main benefits [25, 26]:

- Minimum number of false detected edge pixels.
- Detected edges are very close to the true edges.
- No multiple edge pixels are detected if there is only a single edge point.

Tests have shown that the effectiveness of edge detection strongly depends on the quality of the illumination. The effects of illumination can be compensated by varying the defined thresholds for the edge detection scheme.

The open source C/C++-based library OpenCV [29] has been tested with the aim of reducing the runtime of the edge detection algorithm. Unfortunately, the benefit of a shorter runtime comes together with the lower accuracy in edge detection.

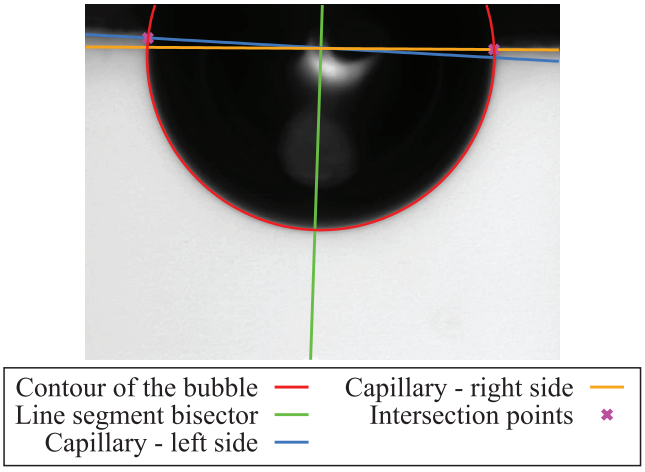


Figure 10. Captured image with detected shapes.

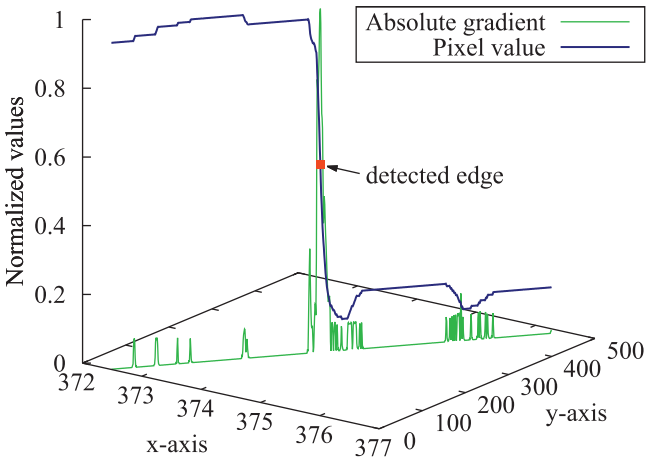


Figure 11. Determination of the cap height by calculating the first derivative along the line segment bisector.

2.3.3. Contour detection. The radii of the capillary tube and the bubble as well as the cap height of the bubble below the capillary must be extracted from the image (see figure 8). The bubble is assumed to be a circle and the edge of the capillary to be a straight line. Following the edge detection, these geometrical shapes have to be localized in the processed image. This is realized by the Hough-transform [24, 30].

Since the built-in MATLAB function did not achieve the desired precision in the determination of the radius and the midpoint, a self-programmed MATLAB script taking into account programming advices given in [31] has been developed to determine the bubble's radius and its midpoint coordinates. Detecting a circle requires the definition of a three-dimensional accumulator array due to the three parameters (two coordinates for the center and one scalar for the radius) needed to define a particular circle [32].

Figure 9 compares the results of the internal MATLAB function and the self programmed MATLAB script for detecting the bubble contour in the captured images. Due to the small amplitude of bubble oscillations, this increase in accuracy is of great importance. Note, that the MATLAB function produces the integer value for the radius, since it cannot interpolate between the pixels. The major disadvantages of the more

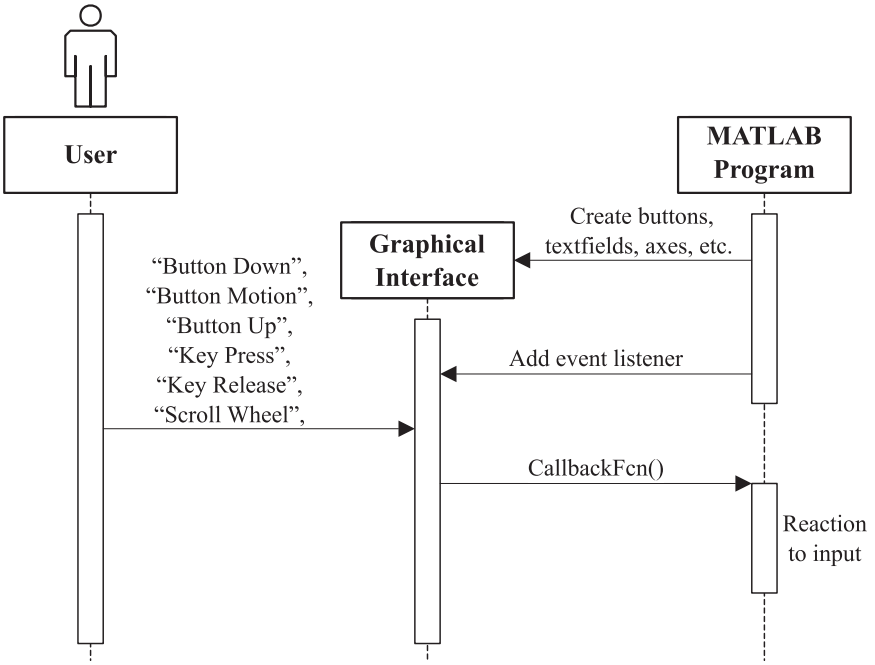


Figure 12. Structure of the MATLAB interface for interaction with the user.

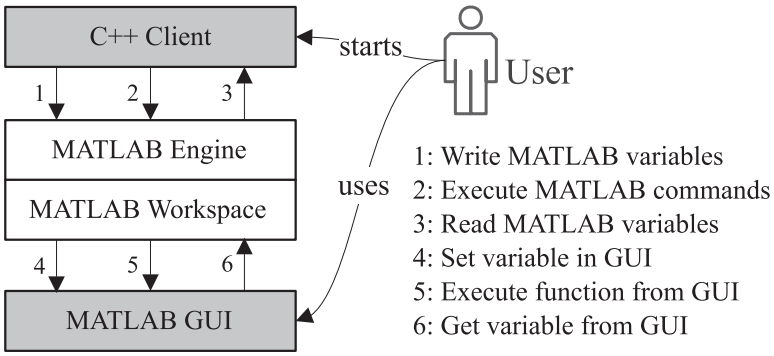


Figure 13. Structure of interlinking MATLAB with C++.

precise algorithm are higher computing effort and longer runtime.

To reduce the calculation time for the contour detection, C++ code is generated from the programmed MATLAB script using the MATLAB Coder Toolbox. This code is executed within MATLAB as MEX-file. Comparing the MATLAB und C++ code in terms of runtime shows, that the C++ code is more than twice faster than the corresponding MATLAB code.

For the ideal hemispherical bubble, the inner radius of the capillary tube R_C is equal both to the radius of the bubble r_0 and the cap height H. This is the setpoint for the automated bubble stabilization. After detecting the bubble contour and the edges of the capillary, the resulting intersection points yield the radius r_0 corresponding to the hemisphere. To account for imperfections of the capillary tip, two lines are necessary to fit the capillary edges. Therefore, a self-programmed MATLAB script based on the Hough transform [24] searches the bottom one-third of the edge detected image for the best-fit lines. Figure 10 illustrates the contour detection scheme.

A major disadvantage of the Hough transform is the large computing effort. Hence, this algorithm can not be employed for determining the cap height of the bubble within a PID-loop. It is used only as the setpoint of the PID-loop (see figure 7).

To determine the bubble size within the PID-loop regulation, the brightness gradient along a straight line perpendicular to the contour of the bubble (green line in figure 10) is examined. This line is located by interpolating the pixel values along the line segment bisector of the intersections between detected lines (blue, orange) and the circle (red). The first derivative along the interpolated line segment bisector is calculated using an internal MATLAB function. The largest value of the first derivative characterizes the edge point of the bubble, which is used to obtain the cap height. Figure 11 illustrates this method, which is approximately 1000 times faster than the Hough transform, but still possesses the same accuracy.

2.4. The graphical user interface

A graphical user interface (GUI) programmed in MATLAB includes live image acquisition, the control of the measurement and visualization of the results. The use of MATLAB as

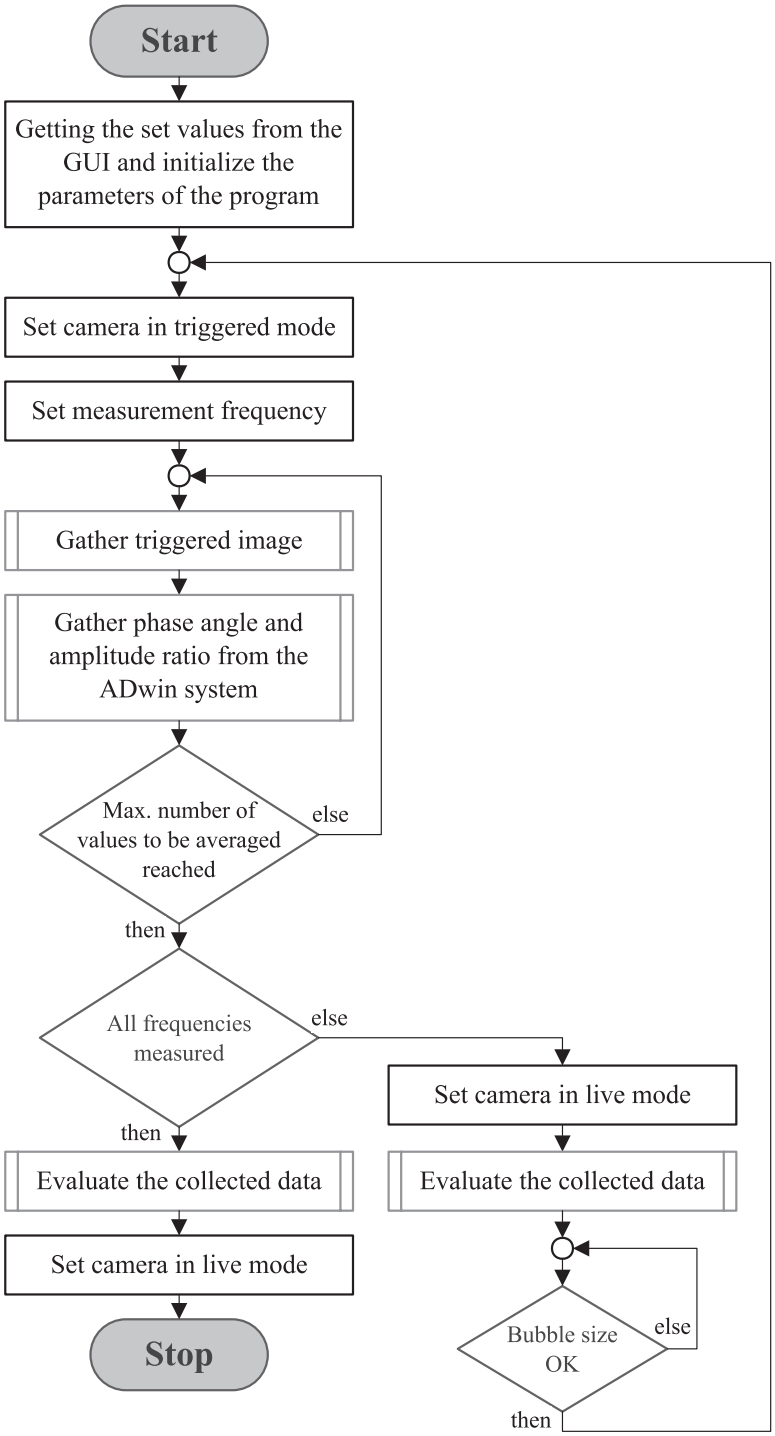


Figure 14. Flowchart representing the structure of the automated measurement within a specified frequency range.

programming language is essentially motivated by variety of integrated functions for data visualization, image processing and the possibility of creating GUIs [33–35]

The diagram in figure 12 illustrates the programming structure behind the developed GUI. A first subprogram creates the graphical elements of the GUI and registers their different event listeners with the individual callback functions. If the user initiates one of the defined events, the corresponding callback function is executed.

The implementation utilizes an object-oriented approach by defining a class structure. For every functional part of the measurement system, an own class is defined which contains objects and methods to realize the desired function.

2.4.1. Implementation of camera control. The used CCD-camera comes with a software development kit (SDK) supporting the programming language Visual C++. The SDK provides functions to detect, enumerate, initialize and grab images from the camera connected via USB [36]. Controlling of the camera via the programmed MATLAB GUI and processing the captured images with MATLAB, requires the interlinking of MATLAB and C++. This functionality

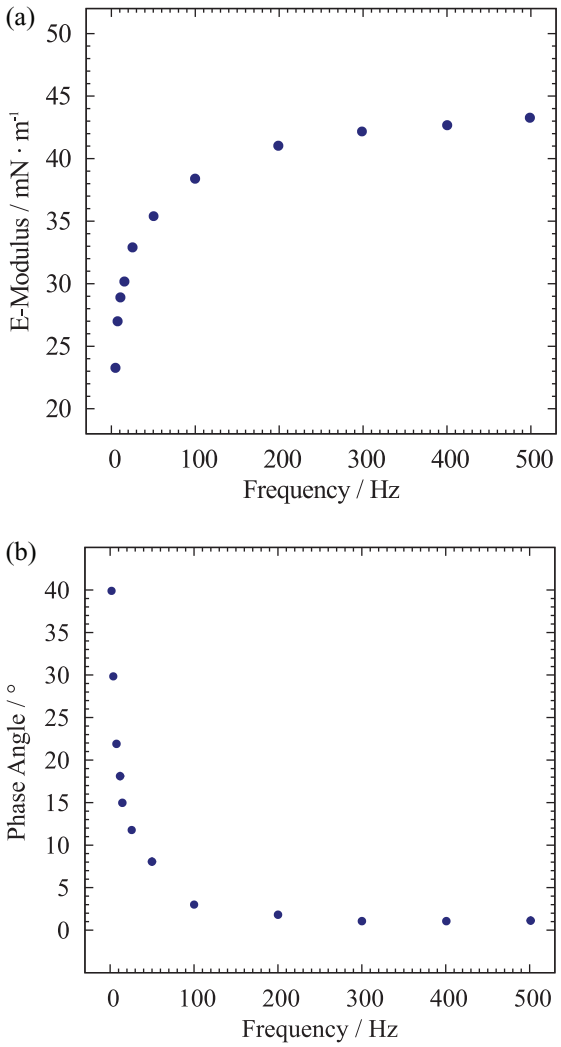


Figure 15. Frequency dependence of the amplitude (a) and the phase (b) of the surface dilational modulus *E*.

is enabled by the so called MATLAB engine [37]. It allows copying variables from the MATLAB engine workspace to the C++ program and vice versa, as well as the execution of MATLAB functions from the C++-program [33]. Figure 13 shows the structure of the realized interlinking of MATLAB with C++ [34].

2.4.2. Measurement automation. The automated determination of the surface dilational modulus *E* is the key feature of the developed oscillating bubble device. Therefore, the section 'Measurement' within the MATLAB GUI contains editable text boxes, as well as buttons to specify a measurement protocol. Adjustable parameters are given by the frequency range, the amplitude of oscillation, the number of measurements to be averaged at each measured frequency and the parameters for automated bubble stabilization. Results from the previously described evaluation of the surface area change and the calculation of the pressure variation and phase shift are combined to determine the complex surface dilational modulus *E*. Figure 14 shows the structure of the implemented functionality. The frequency dependent characteristics of the modulus with its corresponding standard deviations are visualized in both graphical and in tabular form within the GUI. The results can be saved to an ASCII-file.

3. Measurement

Aqueous surfactant solutions are commonly used as detergents. A powerful non-ionic surfactant is tridecyl-dimethylphosphine oxide (C_{13}DMPO). Here, an aqueous solution of C_{13}DMPO with the concentration of 0,125 mmol L^{-1} was characterized by means of the described device in a frequency range from 1 to 500 Hz. The experiment was performed at room temperature. Figures 15(a) and (b) show the absolute value and the phase angle of the complex surface dilational modulus in dependence on the oscillation frequency (frequency sweep), respectively. In both figures, the standard deviation of consecutive measurements in a frequency sweep is less than the symbol size.

Upon increasing the frequency, the absolute value of the modulus approaches a limiting value (see figure 15(a)), whereas the phase angle gradually approaches zero (see figure 15(b)). Such a dependence is usually referred to as surface elastic behavior of an interfacial layer [38]. The device allows for an easy assessment of this frequency dependence, which can be used for characterization of adsorbed layers. Furthermore, surfactant exchange dynamics, i.e. transport properties of surface active species can be determined from this type of measurement [39].

From the practical point of view, the major accomplishment of the present work is that the achieved high degree of automation reduces the measurement time by orders of magnitude in comparison with previously realized experimental setups. The frequency sweep experiment including the filling procedure can be completed in less than 20 min.

4. Conclusion

An oscillating bubble device for measuring the complex surface dilational modulus of aqueous surfactant solutions at the liquid–air interface is developed. The programming tasks required are described in detail.

Essential processes of the device include the generation of a sinusoidal voltage trace driving piezoelectric piston and inducing the harmonic surface area perturbation around the hemispherical reference state of the bubble. Additionally, a TTL-signal acting as the trigger for a CCD-camera to obtain images at maximum and minimum extensions of the bubble is generated. Image processing is applied to retrieve the surface area of the bubble at its extremal states. The amplitude of the resulting pressure response and its phase shift with respect to the induced surface area perturbation are obtained from a lock-in detection scheme based on discretized Fourier components. A PID-loop is used to compensate for potential drifts in bubble size between changes of the oscillation frequency.

Finally, an exemplary measurement at the interface of a non-ionic surfactant solution is conducted to show the capability of the device for determining the frequency dependent characteristics of the absolute value and the phase angle of the

modulus. This can be used to derive transport characteristics and quantify the exchange of material between the bulk phase and the interface of the solution.

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